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when any user -selectable categories are deselected. Details of the deselection should be available on demand.

11.4.5 An alarm should be given when the s pecified cross track limit for deviation from the selected route, if defined by the mariner when route planning, is exceeded.

11.4.6 ECDIS should give a warning or caution or indication as selected by the mariner and related graphical indication if, continuing on its present course and speed, over a specified time or distance set by the mariner, own ship will pass closer than a user -specified distance from a user-selectable category of danger (e.g. obstruction, wreck, rock) that is shallower than the mariner's safety contour or a user -selectable category of aid to navigation. The
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the mariner's safety contour or a user -selectable category of aid to navigation. The user-selectable categories should be the same as user selections for the display of objects and be based on IHO standards. There should be a permanent indicatio n when any of the user-selectable categories are deselected. Details of the deselection should be available on demand.

11.4.7 A graphical indication should be given if the current or the next leg of the selected route passes closer than a user-specified distance from the safety contour.

11.4.8 A graphical indication should be given if the current or the next leg of the selected route goes closer than a user -specified distance from the boundary of a user -selectable category of prohibited ar ea or a geographic area for which special conditions exist (see appendix 4). A graphical indication should also be given if the selected route goes closer than
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appendix 4). A graphical indication should also be given if the selected route goes closer than a user-specified distance from a user-selectable category of point objects, such as a fixed or floating aid to navigation or isolated danger. The user -selectable categories should be the same as user selections for the display of objects and be based on IHO standards.

11.4.9 It should be possible for the mariner to select that the indications of 11. 4.3, 11.4.4, 11.4.6, 11.4.7 and 11.4.8 take into account accuracy information of relevant hydrographic information, as defined by IHO standards.

11.4.10 The ship's position should be derived from a continuous positioning system of an accuracy consistent with the requirements of safe navigation. Whenever possible, a second independent positioning source, preferably of a different type, should be provided. In such cases, ECDIS should be capable of identifying discrepancies between the two sources.

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cases, ECDIS should be capable of identifying discrepancies between the two sources.

11.4.11 ECDIS should provide a warning when the input from position, heading or speed sources is lost. ECDIS should also repeat, but only as an indication, any alerts or indication passed to it from position, heading or speed sources.

11.4.12 A warning should be given by ECDIS when the ship reaches a specified time or distance, set by the mariner, in advance of a critical point on the planned route.

11.4.13 The positioning system and the system database should be on the same geodetic datum. ECDIS should give a warning if this is not the case.

11.4.14 It should be possible to display alternative routes in addition to the selected route. The selected route should be clearly distinguishable from the other routes. During the voyage, it should be possible for the m ariner to modify the selected route or change to an alternative route.

11.4.15 It should be possible to display:

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should be possible for the m ariner to modify the selected route or change to an alternative route.

11.4.15 It should be possible to display:

.1 time labels along a ship's track manually on demand and automatically at intervals selected between 1 and 120 minutes; and

.2 an adequate number of points, free movable electronic bearing lines, variable and fixed range markers and other symbols required for navigation purposes and specified in appendix 3.

11.4.16 It should be possible to enter the geographical coordinates of any position and then display that position on demand. Also, it should be possible to select any point (features, symbol or position) on the display and read its geographical coordinates on demand.

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symbol or position) on the display and read its geographical coordinates on demand.

11.4.17 It should be possible to adjust the displayed geographic position of the ship manually. This manual adjustment should be indicated alphanumerically on the screen, maintained until altered by the mariner and automatically recorded.

11.4.18 ECDIS should provide the capability to enter and plot manually obtained bearing and distance lines of position (LOP), and calculate the resulting position of own ship. It should be possible to use the resulting position as an origin for dead reckoning.

11.4.19 ECDIS should indicate discrepancies between the positions obtained by continuous positioning systems and positions obtained by manual observations.

11.5 Voyage recording

11.5.1 ECDIS should store and be able to reproduce certain minimum elements required to reconstruct the navigation and verify the official database used during the previous 12 hours.

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reconstruct the navigation and verify the official database used during the previous 12 hours. The following data should be recorded at one minute intervals:

.1 to ensure a record of own ship's past track: time, position, heading, and speed;

.2 to ensure a record of official data used: ENC source, edition, date, cell and update history.; and

.3 any changes in safety contour, look ahead and route monitoring alert settings.

11.5.2 ECDIS should output the information listed in 11.5.1.2 and 11.5.1.3 to a voyage data recorder

11.5.3 In addition, ECDIS should record the complete track for the entire voyage, with time marks at intervals not exceeding 4 hours.

11.5.4 It should not be possible to manipulate or change the recorded information.

11.5.5 ECDIS should have a capability to preserve the record of the previous 12 hours and the voyage track.

12 CALCULATIONS AND ACCURACY

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11.5.5 ECDIS should have a capability to preserve the record of the previous 12 hours and the voyage track.

12 CALCULATIONS AND ACCURACY

12.1 The accuracy of all calculations performed by ECDIS should be independent of the characteristics of the output device and should be consistent with the system database accuracy.

12.2 Bearings and distances drawn on the display or those measured between features already drawn on the display should have accuracy no less than that afforded by the resolution of the display.

12.3 The system should be capable of performing and presenting the results of at least the following calculations:

.1 true distance and azimuth between two geographical positions;

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following calculations:

.1 true distance and azimuth between two geographical positions;

.2 geographic position from known position and distance/azimuth; and

.3 geodetic calculations such as spheroidal distance, rhumb line and great circle.

13 PERFORMANCE TESTS, MALFUNCTIONS ALERTS AND INDICATIONS